

EC4215 — Business Econometrics I

Winter 2025 Examination · Paper 1 · Full Marking Scheme

Module	EC4215 — Business Econometrics I
Internal Examiner	Dr Brian Turner
Institution	University College Cork
Session	Winter 2025, Paper 1
Duration	1.5 hours
Structure	Section A — Long Questions: attempt 2 of 3 (100% of the paper). All questions carry equal weight.
Materials	Non-programmable calculator permitted. Formula sheet and statistical tables (t, F, normal, Durbin-Watson) provided.

Coverage note. This scheme answers **all three** Section A questions in full, since you'll choose your best two under exam conditions.

Own-figure rule. This entire paper is calculation-heavy — every t-test, F-test, and diagnostic reading carries forward from an earlier step. An answer that applies the correct method and formula consistently from an earlier arithmetic slip still earns the marks available for the later steps.

Consistency with other sittings. A1(b) and A2(a) here test the same underlying skills (F-test for overall significance; deciding whether to add a variable) as the EC4215 Autumn 2025 scheme, using different numbers. Note specifically that A2(a) on this paper reaches the **opposite** conclusion from the Autumn 2025 scheme's equivalent question — Adjusted R^2 *falls* here when the extra variable is added, rather than rising — so don't assume the direction of the answer carries over between sittings; always recompute.

Source basis. This scheme follows the module's own course material closely throughout (Studenmund, *Using Econometrics*, and Gujarati, *Econometrics by Example*), including the exact steps of the Park, White, and Breusch-Pagan tests for heteroscedasticity and the standard omitted-variable-bias derivation.

Section A · Question 1

Interpreting a rock band earnings regression, and testing a health-insurance regression for significance — 100 marks

A1(a)

[60 marks]

A researcher examines a rock band's earnings: $E_t = 44.3 + 0.15D_t + 7.7T_t + 56.1A_t$, standard errors (0.04), (2.2), (37.2), $\bar{R}^2=0.78$, $n=24$. Where E_t = earnings (€000), D_t = song downloads (000s), T_t = concert tickets sold (000s), $A_t = 1$ if a new album was released, 0 otherwise. (i) Carefully interpret the above findings. (ii) Does the above equation fit your expectations? Why or why not? (iii) If you could add one more variable into the above regression, what would it be and why?

A1(b)

[40 marks]

A researcher examines private health insurance take-up in Ireland, Q1 2003 to Q4 2016: $T_t = 784.3469 + 0.040072C_t + 9.218391P_t - 1.593816I_t$, standard errors (0.008084), (10.58059), (0.707827), $\bar{R}^2=0.56833$, $n=56$, $RSS=76,986.76$, $ESS=111,648.9$. (i) Looking at this output (without carrying out any statistical tests), would you expect the regression to be statistically significant? Explain your answer. (ii) Formally test this regression for statistical significance at the 5% level of significance. Explain the steps you take.

FULL MODEL ANSWER

(i) Interpreting each coefficient, ceteris paribus:

Variable	Coefficient	t-statistic	Interpretation
Intercept	44.3	—	The baseline earnings (€44,300) if downloads, ticket sales, and album releases were all zero — a fixed income component (e.g. existing royalties, other revenue) unrelated to current-year activity
			A 1,000-unit rise in song downloads is associated with a €150 rise in earnings (0.15

D_t (downloads, 000s)	0.15	3.75	× €1,000, since E is in €000), holding tickets sold and album release constant — a modest but statistically significant per-download revenue contribution, consistent with typical streaming/download royalty rates
T_t (tickets, 000s)	7.7	3.50	A 1,000-unit rise in concert tickets sold is associated with a €7,700 rise in earnings, holding downloads and album release constant — a much larger effect than downloads, consistent with concert ticket revenue per unit being far higher than a single download royalty
A_t (album dummy)	56.1	1.508	Releasing a new album is associated with €56,100 higher earnings than not releasing one, holding downloads and tickets constant — but this coefficient is imprecisely estimated (large standard error) and is NOT statistically significant, so this large point estimate cannot be confidently distinguished from zero

Statistical significance (t-tests, $df = n - k - 1 = 24 - 3 - 1 = 20$): comparing each t-statistic to the critical value from the appendix t-table at $df=20$.

Variable	t-statistic	Critical t (5%, two-tail, $df=20$)	Significant?
D_t	3.75	2.086	Yes
T_t	3.50	2.086	Yes
A_t	1.508	2.086	No — not even significant at the 10% two-tail level (critical t = 1.725)

$\bar{R}^2 = 0.78$: 78% of the variation in the band's earnings is explained by the model, after adjusting for the number of explanatory variables — a strong fit for this kind of revenue equation.

Mark Allocation — Part (a)(i) (35% of Q. A1(a))

Tier	Marks	What separates this tier
Full credit	76-100%	All four coefficients correctly interpreted in ceteris paribus, economically-meaningful magnitude terms AND all three t-tests correctly calculated and compared to the correct critical value at the correct df AND the album dummy's lack of significance explicitly flagged despite its large point estimate AND $\bar{R}^2$ correctly interpreted.
Good	51-75%	Coefficients correctly interpreted with at least two of the three t-tests correctly carried out.
Partial	26-50%	Coefficients interpreted only by sign/direction without magnitude language or significance testing.
Minimal	0-25%	No credible interpretation given.

(ii) Does the equation fit expectations? Largely yes for the two significant variables, with a genuine caveat on the album dummy.
Downloads and ticket sales both carry the expected positive sign, and the relative magnitude — ticket sales having a far larger per-unit effect than downloads — matches the economic intuition that a concert ticket generates

far more revenue per unit sold than a single song download. The album-release dummy's positive point estimate also matches the plausible expectation that a new album boosts earnings (both directly, and by driving ticket demand and further downloads), but the fact that this coefficient is **not statistically significant** is somewhat disappointing given how plausible the underlying story is — with only $n=24$ observations, the standard error on a dummy variable capturing a relatively infrequent event (an album release) is often large, and this may simply reflect insufficient data (few album-release years in the sample) rather than genuine absence of an effect.

Mark Allocation — Part (a)(ii) (15% of Q. A1(a))

Tier	Marks	What separates this tier
Full credit	76-100%	Correctly identifies downloads and tickets as matching expectations with the relative-magnitude reasoning AND correctly engages with the album dummy's plausible-but-insignificant result, offering a credible reason (small sample, infrequent event) for the imprecision.
Good	51-75%	A reasonable overall verdict given with some engagement with the album dummy's significance issue.
Partial	26-50%	States the equation "fits expectations" without engaging each variable individually or addressing the significance issue.
Minimal	0-25%	No credible answer given.

(iii) An additional variable worth adding: streaming revenue/play counts (e.g. Spotify or equivalent platform streams) would be a strong candidate. Downloads (D_t) capture only one, increasingly less dominant, form of recorded-music consumption — streaming has become the primary revenue channel for most artists in recent years, and is economically distinct from a one-off download purchase (streaming royalties accrue per play, not per purchase). Omitting streaming activity risks omitted variable bias if streaming numbers are correlated with the included variables (e.g. a popular album likely drives both downloads and streams together), and including it directly would let the model capture a major, increasingly important revenue source in its own right.

EXAM TECHNIQUE

Other well-justified answers — average ticket price (since T_t only counts tickets sold, not the price charged, so two years with identical ticket counts but very different pricing would look identical to the model), merchandise sales, or a social-media following/engagement metric as a modern demand proxy — are equally creditable provided the economic reasoning is clearly given.

Mark Allocation — Part (a)(iii) (10% of Q. A1(a))

Tier	Marks	What separates this tier
Full credit	76-100%	A specific, economically well-motivated additional variable proposed with clear reasoning for why it should affect earnings and why its omission is a concern.
Good	51-75%	A reasonable variable proposed with some reasoning given.
Partial	26-50%	A variable named without economic reasoning.
Minimal	0-25%	No credible variable proposed.

(b)(i) Would you expect the regression to be significant, without testing? Yes, likely. $\bar{R}^2=0.568$ represents a solid share of variation explained, and with $n=56$ observations against only $k=3$ explanatory variables,

there are 52 degrees of freedom — a large sample relative to the number of parameters being estimated. This combination (a respectable $\bar{R}^2$ with ample degrees of freedom) typically translates into a statistically significant F-statistic for overall significance.

Mark Allocation — Part (b)(i) (10% of Q. A1(b))

Tier	Marks	What separates this tier
Full credit	76-100%	Correctly predicts significance AND correctly reasons from both the $\bar{R}^2$ value and the large sample size relative to k.
Good	51-75%	Correct prediction with some reasoning given.
Partial	26-50%	A prediction given based only on the $\bar{R}^2$ value.
Minimal	0-25%	No credible answer given.

(b)(ii) Formal F-test for overall significance: $H_0: \beta_1=\beta_2=\beta_3=0$ (none of C, P, I jointly explain take-up), against H_a : at least one $\beta_k \neq 0$.

Step	Calculation	Result
TSS = RSS + ESS	76,986.76 + 111,648.9	188,635.66
$F = (ESS/k) / (RSS/(n-k-1))$	$(111,648.9/3) / (76,986.76/52)$	25.137
Degrees of freedom	$df_1=k=3, df_2=n-k-1=52$	F(3,52)
Critical F(3,52) at 5%	interpolated between F(3,40)=2.84 and F(3,60)=2.76	≈ 2.79

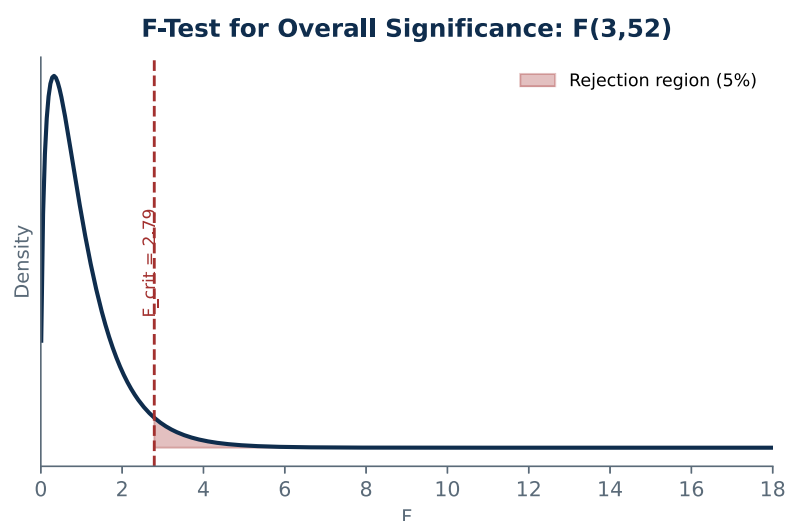


Fig. 1 — The calculated F-statistic (25.14) falls decisively into the rejection region beyond the critical value (≈ 2.79).

The steps, and the conclusion: (1) state H_0 and H_a ; (2) calculate F using the given ESS and RSS; (3) identify $df_1=k$ and $df_2=n-k-1$; (4) find the critical F-value at 5% for those degrees of freedom; (5) compare. Since the calculated F (25.137) is far larger than the critical value (≈ 2.79), we **reject the null hypothesis** — the regression **is statistically significant overall** at the 5% level, confirming the prediction in Part (b)(i).

Mark Allocation — Part (b)(ii) (30% of Q. A1(b))

Tier	Marks	What separates this tier
Full credit	76-100%	Correct null/alternative hypotheses stated AND F correctly

calculated as 25.137 **AND** correct degrees of freedom identified (3, 52) **AND** correctly compared to the (interpolated) critical value **AND** the correct "reject H0, significant" conclusion given.

Good	51-75%	F correctly calculated and degrees of freedom correct, but the critical value comparison or conclusion has an error.
Partial	26-50%	F formula set up correctly but arithmetic error in the final statistic.
Minimal	0-25%	No coherent F-test attempted.

Question A1 Total: 100%

Section A · Question 2

Comparing two nested cinema-attendance regressions, and the general consequences of omitting a relevant variable — 100 marks

A2(a)

[70 marks]

A researcher examining cinema attendance gets: $CT_t = 52.1 + 9.6RF_t + 17.3AD_t + 9.2SH_t$ (Eqn. 2.1), SEs (2.8), (5.4), (3.1), $N=52$, Adjusted $R^2=0.674$. Adding a major-awards-ceremony dummy gives: $CT_t = 51.4 + 9.5RF_t + 17.2AD_t + 9.0SH_t + 2.4AC_t$ (Eqn. 2.2), SEs (3.0), (5.9), (3.3), (1.9), $N=52$, Adjusted $R^2=0.643$. Should the researcher include AC_t (i.e. go with Eqn. 2.2 instead of Eqn. 2.1)? Explain your answer.

A2(b)

[30 marks]

Suppose that the 'true' regression model is Eqn. 2.1: $Y_i = b_0 + b_1X_{1i} + b_2X_{2i} + u_i$, where u_i is the classical disturbance term. However, suppose a researcher estimates Eqn. 2.2: $Y_i = a_0 + a_1X_{1i} + v_i$. Assess the consequences of estimating the mis-specified model (Eqn. 2.2) instead of the 'true' model (Eqn. 2.1).

FULL MODEL ANSWER

The three pieces of evidence to weigh: as with any variable-inclusion decision, the change in explanatory power, the added variable's own statistical significance, and the stability of the other coefficients across specifications should all be considered together.

1. Adjusted R^2 FALLS when AC is added: unlike plain R^2 , which mechanically rises whenever any variable is added regardless of its relevance, Adjusted R^2 penalises the addition of variables that don't genuinely improve

the model. Here, Adjusted R^2 **falls** from 0.674 to 0.643 when AC is added — a clear signal that AC is not pulling its weight, and that the extra parameter being estimated is costing more in lost degrees of freedom than it's adding in genuine explanatory power.

2. AC is NOT individually statistically significant: testing $H_0: \beta(AC)=0$ with $t = 2.4/1.9 = \mathbf{1.263}$, against a critical $t(0.05, df=52-4-1=47, \text{two-tail}) \approx 2.012$ — since $1.263 < 2.012$, **AC is not statistically significant** at the 5% level. Its true coefficient cannot be confidently distinguished from zero.

3. The other coefficients remain stable across both specifications: RF (9.6→9.5), AD (17.3→17.2), and SH (9.2→9.0) all stay close in magnitude and keep their expected signs across the two equations — this stability further suggests Eqn. 2.1 was not suffering from meaningful omitted variable bias by excluding AC in the first place.

Conclusion: No, the researcher should NOT include AC_t — Eqn. 2.1 (without the awards-ceremony dummy) is the better specification. All three pieces of evidence point the same direction: the fall in Adjusted R^2 , AC's own statistical insignificance, and the stability of the other coefficients without it, together indicate that whether a major awards ceremony falls in a given week does not have a detectable, independent effect on cinema attendance once rainfall, advertising spend, and school holidays are already controlled for.

EXAM TECHNIQUE

Compare this directly with the EC4215 Autumn 2025 scheme's Q. A2(a), which asks the same type of question but reaches the *opposite* conclusion (Adjusted R^2 rises there, so the variable should be included) — the diagnostic method is identical in both cases; only the direction the numbers point in differs. Always recompute rather than pattern-matching to a previous answer.

Mark Allocation — Part (a) (70%)

Tier	Marks	What separates this tier
Full credit	76-100%	Correctly identifies the FALL in Adjusted R^2 as evidence against inclusion, explaining why this is informative AND correctly calculates AC's t-statistic (1.263) and compares to the correct critical value (≈ 2.012) AND notes the stability of the other coefficients AND reaches the correct "do NOT include AC, stick with Eqn. 2.1" conclusion.
Good	51-75%	Correctly uses the Adjusted R^2 fall and/or AC's insignificance to recommend exclusion, but doesn't develop both pieces of evidence together.
Partial	26-50%	States a conclusion based only on the Adjusted R^2 change, without testing AC's individual significance.
Minimal	0-25%	No credible comparison of the two equations given, or the conclusion is reversed (recommending inclusion despite the evidence).

Setting up the derivation: since the true model includes X_{2i} but the estimated model omits it, the true relationship can be rewritten by substituting the true model into what the misspecified model is actually estimating: $v_i = \beta_2 X_{2i} + u_i$ — the omitted variable's entire effect, along with the original disturbance, is absorbed into the misspecified model's error term.

Case 1 — X_1 and X_2 are correlated (the typical, more serious case)

The consequence: $\hat{\alpha}_1$ is biased and inconsistent. if X_1 and X_2 are correlated, then the new error term v_i (which contains $\beta_2 X_{2i}$) is also correlated with X_{1i} — directly violating the classical assumption that the regressor is uncorrelated with the error term. This means:

- $E[\hat{\alpha}_1] = \beta_1 + \beta_2 \delta_1$, where δ_1 is the coefficient from a hypothetical auxiliary regression of X_2 on X_1 — the estimator is systematically off from the true β_1 by an amount equal to this **expected bias** term.
- The **direction** of the bias is determined by the sign of β_2 (the omitted variable's true effect) multiplied by the sign of δ_1 (the relationship between the omitted and included variable) — exactly the expected-bias-analysis formula used elsewhere in this module (see the EC4215 Autumn 2025 scheme, Q. A2(b)).
- This bias does **not** disappear as the sample size grows — α_1 remains **inconsistent** as well as biased, since the correlation between X_1 and the error term persists regardless of n .
- Hypothesis tests and confidence intervals built on α_1 become **unreliable**, since they are centred on a systematically wrong value rather than the true β_1 .

Case 2 — X_1 and X_2 are uncorrelated

The consequence: $\hat{\alpha}_1$ remains unbiased, but is less efficient. if X_1 and X_2 happen to be uncorrelated, then $\delta_1=0$, so the bias term $\beta_2 \delta_1$ vanishes entirely — **$\hat{\alpha}_1$ remains an unbiased estimator of β_1** even though X_2 is omitted. However, the omitted variable's genuine explanatory power is still lost, and $\hat{\alpha}$ gets absorbed entirely into the error term — **inflating the error variance** (σ^2 is larger than it should be) and, correspondingly, **inflating the standard error of $\hat{\alpha}_1$** . The estimator is still centred correctly on average, but is **less precise** (higher variance, lower power in hypothesis tests) than it would have been had X_2 been correctly included.

Summary: omitting a genuinely relevant variable is unambiguously costly — in the (more common, more serious) case where the omitted and included variables are correlated, the remaining coefficient becomes biased and inconsistent, with unreliable hypothesis tests; even in the more benign case of no correlation between them, the omission still costs precision by inflating the estimator's standard error, even though the point estimate itself remains centred correctly.

Mark Allocation — Part (b) (30%)

Tier	Marks	What separates this tier
Full credit	76-100%	Correctly derives that the omitted variable's effect is absorbed into the error term of the misspecified model AND correctly distinguishes the correlated case (bias = $\beta_2 \delta_1$, biased AND inconsistent, unreliable hypothesis tests) from the uncorrelated case (unbiased but less efficient, inflated variance/SE) AND gives a clear overall summary.
Good	51-75%	Correctly identifies the biased/inconsistent consequence in the correlated case, but the uncorrelated case (remains unbiased, only loses efficiency) is missing or incorrect.
Partial	26-50%	States that omitting X_2 "causes bias" without the correlation-dependent nuance or the bias formula.
Minimal	0-25%	No credible assessment given.

Question A2 Total: 100%

Section A · Question 3

Diagnosing an EViews regression output, the White test, and comparing heteroscedasticity tests — 100 marks

A3(a)

[40 marks]

A researcher regresses HEXPTOT (total health expenditure) on GDP, POP75PROP, ALOS, and NONSMOK across European countries ($n=26$). The EViews output shows: $C=36413.90$ ($t=0.7477$, $p=0.4630$); $GDP=0.108243$ ($t=33.327$, $p=0.0000$); $POP75PROP=-650.1453$ ($t=-0.3384$, $p=0.7384$); $ALOS=1534.125$ ($t=0.4599$, $p=0.6503$); $NONSMOK=-600.0428$ ($t=-0.9559$, $p=0.3500$). $R^2=0.986665$, Adjusted $R^2=0.984126$, F-statistic= 388.4654 ($p=0.000000$), Durbin-Watson stat= 2.580435 . (i) Does anything in the above results suggest that the error term in this regression suffers from any econometric problems? Explain your answer. (ii) The researcher then ran the White Test and obtained a Chi-squared statistic of 25.50482 , compared with a critical Chi-squared statistic (for 14 degrees of freedom with a 5% level of significance) of 23.7 . Does this result suggest the presence of heteroscedasticity? Explain your answer.

A3(b)

[60 marks]

Describe at least two different tests for detecting heteroscedasticity, and outline the similarities and differences between them.

FULL MODEL ANSWER

(i) The diagnostic pattern to notice: the overall regression is **extremely strong** — $R^2=0.987$, Adjusted $R^2=0.984$, and the overall F-test is overwhelmingly significant ($F=388.47$, $p=0.000000$) — yet, looking at the individual coefficients, only **one** of the four explanatory variables (GDP) is individually statistically significant ($p=0.0000$); **POP75PROP, ALOS, and NONSMOK are all highly insignificant** ($p=0.7384$, 0.6503 , and 0.3500 respectively — all far above any conventional significance threshold), and even the intercept is insignificant ($p=0.4630$).

Variable	Coefficient	t-statistic	p-value	Individually significant?
C	36,413.90	0.748	0.4630	No
GDP	0.108243	33.327	0.0000	Yes — overwhelmingly
POP75PROP	-650.1453	-0.338	0.7384	No
ALOS	1,534.125	0.460	0.6503	No
NONSMOK	-600.0428	-0.956	0.3500	No

What this pattern indicates: multicollinearity. the combination of **very high overall explanatory power and joint significance, alongside individually insignificant coefficients** on almost every variable, is the textbook signature of **multicollinearity** — the explanatory variables are highly correlated with one another (plausibly here: countries with larger GDP likely also have more hospital beds, different population age structures, and different health system characteristics that move together). When regressors are highly correlated, the model can still explain the dependent variable very well *jointly*, but it becomes statistically very difficult to isolate each variable's *separate*, independent contribution — this inflates the standard errors of the individual coefficients (widening confidence intervals and shrinking t-statistics toward insignificance) without necessarily biasing the coefficient point estimates themselves or harming the model's overall predictive fit.

A secondary check — the Durbin-Watson statistic: $DW=2.580$ is somewhat above 2, which could hint at negative autocorrelation, but checking the formal decision rule ($n=26$, $k=4$: $dL=1.143$, $dU=1.652$, so $4-dU=2.348$ and $4-dL=2.857$) shows $d=2.580$ falls in the **inconclusive** zone for negative autocorrelation ($2.348 < 2.580 < 2.857$) — this is not a clear-cut diagnosis either way, and is a secondary consideration well behind the much more obvious multicollinearity signature above.

Mark Allocation — Part (a)(i) (20%)

Tier	Marks	What separates this tier
Full credit	76-100%	Correctly identifies the high-overall-significance/low-individual-significance pattern AND correctly diagnoses multicollinearity as the underlying problem, with the economic reasoning for why the regressors are likely correlated with each other AND notes the DW statistic is inconclusive as a secondary, less clear-cut observation.
Good	51-75%	Correctly diagnoses multicollinearity from the significance pattern, without engaging the DW statistic or the underlying economic reasoning.
Partial	26-50%	Notices that "something is odd" about the coefficients without correctly naming multicollinearity.
Minimal	0-25%	No credible diagnosis given.

(ii) Does the White test result indicate heteroscedasticity? Yes. The White test statistic (NR^2 , following a chi-square distribution under the null of homoscedasticity) is **25.50482**, compared against a critical chi-square value of **23.7** at 14 degrees of freedom and the 5% significance level. Since the calculated test statistic (25.50) **exceeds** the critical value (23.7), we **reject the null hypothesis of homoscedasticity** — the White test provides statistical evidence that **heteroscedasticity is present** in this regression's error term.

Mark Allocation — Part (a)(ii) (20%)

Tier	Marks	What separates this tier
Full credit	76-100%	Correctly compares the test statistic to the critical value AND correctly concludes heteroscedasticity is present, with the reasoning (reject H_0 of homoscedasticity since test statistic exceeds critical value) explicitly stated.
Good	51-75%	Correct conclusion given with the comparison shown but not fully explained.

Partial **26-50%** States a conclusion without clearly comparing the two values.

Minimal **0-25%** Incorrect conclusion given, or no credible answer.

Test 1 — The White Test: (1) estimate the original OLS regression and obtain the residuals; (2) regress the **squared** residuals on all the original regressors, their **squared terms**, and the **pairwise cross-products** of each regressor with every other regressor; (3) the test statistic is **NR²** (sample size $\times$ the R^2 from this auxiliary regression), which follows a chi-square distribution with degrees of freedom equal to the number of slope coefficients estimated in the auxiliary regression; (4) if NR^2 exceeds the critical chi-square value, reject the null of homoscedasticity (exactly as applied in Part (a)(ii) above).

Test 2 — The Breusch-Pagan (BP) Test: (1) estimate the original OLS regression and obtain the **squared** residuals; (2) regress the squared residuals on the **original k regressors only** (no squared terms or cross-products); (3) the test statistic is the **F-statistic** from this auxiliary regression, with $(k-1, n-k)$ degrees of freedom; (4) if the computed F-statistic is statistically significant, reject the null hypothesis that all the slope coefficients in the auxiliary regression are jointly zero (i.e. reject homoscedasticity).

Similarities and differences

	White Test	Breusch-Pagan Test
Starting point	OLS residuals from the original regression, squared	OLS residuals from the original regression, squared
Auxiliary regressors used	Original regressors, their squares, AND all pairwise cross-products	Original regressors ONLY
Test statistic	$NR^2 \sim$ chi-square, df = number of auxiliary slope coefficients	F-statistic, $df = (k-1, n-k)$
Null hypothesis	Homoscedasticity (auxiliary R^2 is essentially zero)	Homoscedasticity (all auxiliary slope coefficients are zero)
Generality	More general — does not assume a specific functional form for how variance depends on the regressors	Assumes error variance is a linear function of the original regressors specifically
Cost of that generality	Uses many more degrees of freedom (squares and cross-products), which can reduce power, especially with several regressors and a modest sample	More parsimonious — fewer auxiliary regressors, generally more power if its linear-form assumption is roughly correct

The shared logic underlying both: both tests follow the same basic two-step structure — estimate the original model, then test whether the *squared* residuals can be systematically predicted by some set of variables. If they can, that's direct evidence the error variance isn't constant (i.e. it depends on those variables), which is exactly what heteroscedasticity means; if they can't, there's no such evidence and homoscedasticity cannot be rejected. The key difference is simply *how much* and *what kind* of structure each test allows the variance to depend on — White's test is deliberately unrestrictive about the functional form (at the cost of degrees of freedom), while Breusch-Pagan is more targeted and parsimonious, but only valid if its assumption that variance is linear in the original regressors happens to be roughly right.

EXAM TECHNIQUE

The Park test is a third valid alternative if a specific answer requires more than two — it differs from both White and Breusch-Pagan in requiring the researcher to choose a single proportionality factor Z in advance (rather than using the regressors automatically), tested with a simple t-test on Z 's coefficient in a regression of the log of squared residuals on $\log(Z)$. Its strength (a clear, interpretable link to a chosen factor) is also its weakness (Z must be correctly guessed in advance, or the test can miss genuine heteroscedasticity driven by a different factor).

Mark Allocation — Part (b) (60%)

Tier	Marks	What separates this tier
Full credit	76-100%	Both tests correctly described with their specific steps (auxiliary regression construction, test statistic, distribution, decision rule) AND at least three genuine similarities/differences correctly identified and explained (shared squared-residual starting point; White's use of squares/cross-products vs. BP's original-regressors-only; chi-square vs. F test statistic; White's greater generality vs. BP's greater power/parsimony).
Good	51-75%	Both tests correctly described, with at least one clear similarity and one clear difference identified.
Partial	26-50%	Only one test correctly and fully described.
Minimal	0-25%	No credible tests described.

Question A3 Total: 100%